

How can OR help to deliver safe AI?

The OR Society and Validate AI hit the road to find out....

The journey toward trustworthy AI continues to evolve amid global discussions and collaborative initiatives.

In November, the first global AI Safety Summit was hosted in the UK, bringing together AI experts from 28 countries advocating responsible AI evolution. One key outcome was a declaration by leading AI nations to collaborate on the opportunities AI brings and identify and regulate AI risks. Additionally, the establishment of the UK's AI Safety Institute marked a significant step forward in addressing the potential hazards of advanced AI models.

However, the groundwork for enhancing trust in AI is already in progress. The OR Society and Validate AI, a leading community interest group focused on AI trustworthiness and safety, have partnered to host a series of workshops to advance education and debate on methods and solutions to build and embed trust and safety within AI systems.

From this Autumn and throughout 2024 in the UK and internationally, the workshops will be hosted by Imperial College London, The University of Edinburgh, University of Southampton, and Vienna University of Economics and Business, as well as renowned societies like the British Computer Society. They will take a deep dive into the use of AI in diverse sectors including healthcare, insurance, finance, taxation, gaming, and robotics and explore AI implementation challenges associated, including security and procurement.

Invaluable insights:

The most recent event, hosted by University of Southampton, gathered experts from industry, academia, and government to consider how operational research can be used to validate and embed trustworthy AI.

Host Martin Kunc, Professor of Management Science at Southampton Business School explained the University of Southampton is playing a central role in developing AI in the UK and has received substantial funding to drive innovation including the establishment of a £15m training centre to nurture British tech talent and to develop AI solutions to tackle climate change.

Shakeel Khan, co-founder and CEO of Validate AI, introduced the three key aims of the work of Validate AI: to develop practitioner centric standards to validate AI system; drive Innovation through research and development to address practitioner problems and; to bring together a community at conferences and workshops to cross-fertilise ideas.

While Seb Hargreaves, Executive Director at The OR Society highlighted The OR Society's history of supporting technological innovation, which is why sponsoring these educational AI workshops were so key and he highlighted the outcomes of this OR session would be of particular interest to members.

The experts explained the goals of the workshop:

- To facilitate panel and workshop discussion to identify the key issues and solutions.
- Capture diverse specialist views: Process, Business, Analytics, Policy challenges for OR enabled AI.
- Contribute to a first draft of a paper to share innovation in OR Enabled AI approaches.
- Identify potential projects to pilot in the workplace and develop our framework.

It is also hoped the ideas and outputs will also serve as a catalyst to promote a movement for the adoption of relevant OR methods.

The Panel Debate

At the panel discussion, chaired by Prof. Martin Kunc, experts from different industries shared their experience and views on the challenges of embedding AI.

John Lord, an analyst at HM Revenue and Customs, emphasised the importance of tailoring AI solutions to address specific issues and highlighted that defining the specific business problem AI should solve must be the starting point. He said a key challenge in procuring AI systems is the limited information available on how they are built.

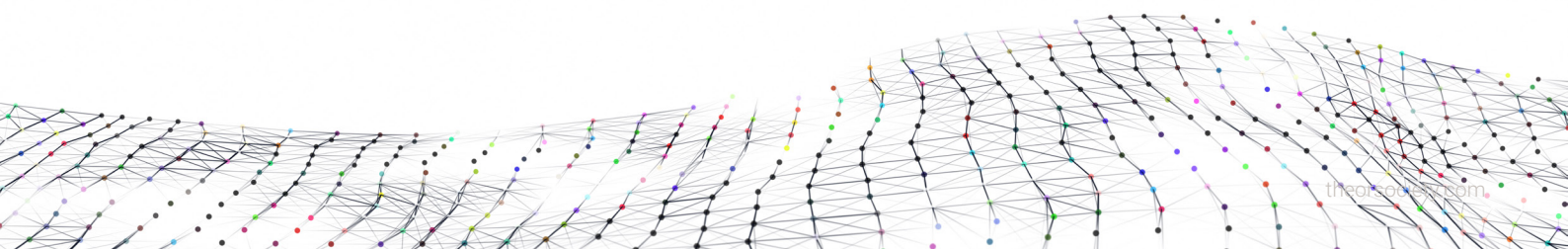
Penny Jones, Ethical Governance Lead at Zurich Insurance, pointed out that at Zurich while machine learning is used to process volumes of information, human oversight is critical particularly in supporting key decisions about insurance claims. Any decision about AI is first based on how a customer would feel about AI being used in a situation.

Rob Claxton, a Senior Manager in Applied Research at British Telecom, discussed the need for consistency in building AI systems, emphasising that consistency would boost trust in AI.

Bethan Freeland a Senior Engineer in Machine Learning and Artificial Intelligence at Roke said a key challenge was that AI's rapid pace is moving more quickly than regulatory frameworks. She emphasised the need for comprehensive departmental involvement before AI implementation to bridge this gap. She also echoed the need for human oversight particularly when AI or machines were used to assist in military operations.

How could OR help deliver AI in a more trusted manner?

The experts agreed that OR methods can be applied to improve AI validation at many different stages – from scoping the solution, to testing and implementation. However, humans need to be part of the decision-making process. In any military situation, or medical situation, humans must make final decisions until you can be 100% confident data is accurate. While machines can process volumes of data quickly, humans are good at looking at more limited information and making decisions and critically, understanding the



context needed in decision making.

All agreed soft OR techniques can be used to ensure people were having the right conversations and asking the right questions about AI use and ensure everyone understands the processes. Hard OR could be deployed in the testing, modelling, and learning phases of deployment to ensure greater success and create 'what if' scenarios.

Practical Applications: Bridging AI assurance with Operational Research

The latter part of the workshop engaged delegates in an insightful brainstorming session, where six groups analysed and ranked crucial actions at each stage of Validate AI's AI Assurance Certification checklist. These discussions sought to identify applicable soft and hard OR techniques pivotal in enhancing implementation processes for increased safety assurance in AI applications.

The breakdown of stages and findings proved enlightening:

Scoping the Problem: The group prioritised defining customer needs and ensuring a clear understanding before conducting feasibility studies. They suggested using various soft OR techniques like process mapping, causal loop diagrams, multiple-criteria decision-making (MCDM), and war gaming techniques to bolster planning and project success.

Data Preparation: The group emphasised the importance of clearly defining data usage and recommended employing machine learning, fuzzy matching, Monte Carlo simulations, and process mapping to address biases and determine data impacts on business and stakeholders.

Algorithmic Development: The focus was on validation methods and problem characterisation. Soft OR techniques such as process mapping were highlighted to quantify costs and explore model options.

Deployment and Maintenance: Testing performance, transparency in AI actions, and continuous monitoring were stressed. OR techniques like simulations, scenario planning, and process mapping were recommended to aid in these actions.

Data Governance: Identifying data locations, ensuring protection, and compliance with regulations were key actions. Stakeholder management and process mapping were proposed as critical soft OR techniques for streamlined governance.

Ethical Considerations: Addressing sub-group harm, bias testing, and ensuring accountability were the focal points. OR techniques, including simulation for bias checking and process mapping for problem structuring, were highlighted as solutions.

The workshop's brainstorming sessions uncovered significant insights. From scoping problems to addressing ethical

considerations, OR techniques emerged as key facilitators in enhancing AI implementation processes. Stakeholders' buy-in, problem structuring, and bias-checking simulations were highlighted as critical aspects.

Summing up the session, Shakeel Khan said, "Validate AI was delighted to work with Southampton University to deliver the OR Enabled AI workshop. The levels of ideas generated exceeded our expectations on what was a difficult topic, as to the potential impact OR to boost trustworthiness in AI delivery. We look forward to working with experts who attended the workshop to further develop the concept of OR Enabled AI.

His views were echoed by Professor Kunc who said, "The workshop is another step on the extensive contribution of the University of Southampton to creating a safe and trustworthy AI. We were pleased that members of the Faculty of Social Science and CORMSIS coordinated the preparation of the workshop. I found the discussion about the use of OR in the project cycle of AI solution very stimulating and with a lot of potential for the future development of the OR field and the community."

Delegates were also enthusiastic. John Lord from HMRC said, "The session brought analysts and users together to understand where OR techniques could help implement AI. It was great to hear views from defence, health, tax, insurance, etc., demonstrating clearly that context is key when deploying models such as considering worst case scenarios in health or military scenarios."

While Penny Jones said, "What a fantastic afternoon discussing such an important topic. I think we all agree that we could have carried on for several more hours".

Mithu Norris from the Government OR Central Management Unit hailed the workshop for its insightful discussions across public and private sectors and for sparking conversations about AI's limitations and future trajectories.

It is clear from the insights and views shared by experts at the workshop that OR has a vital role to play as we move towards safe and responsible AI deployment.

The next steps are now to write a white paper to consolidate all the ideas and outputs from the workshop to influence operational processes to deliver trusted, safe and responsible AI.

There is also the possibility of developing a learning curriculum on operationalising AI.

Please get in touch with Shakeel at: shakeel.khan@validateai.org if you would be keen to get involved. 

